

Project Report: IE500 Data Mining

Ethnic Bias and Generalization in Facial Attractiveness Prediction from Geometric Beauty Markers

Team Number: 3

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1 Application Area and Goals

From surgical planning to recommendation systems, facial attractiveness prediction is increasingly deployed in settings where its outputs shape real decisions about people. Unlike most computer vision tasks, attractiveness ratings are inherently subjective and culturally shaped: what is considered attractive varies across ethnic groups and geographies. Models trained on data from a narrow demographic may learn culture-specific beauty standards rather than broadly applicable visual patterns, raising both methodological and ethical concerns.

To understand which facial proportions drive attractiveness ratings and whether these patterns vary across ethnicities, we extract 30 geometric ratios and 9 expression features from MediaPipe [1] landmarks and compare several regression approaches. Named features allow direct interpretation of model behavior, unlike pixel-level deep learning models where individual predictions remain opaque.

Our core research questions are:

1. How well do models generalize across datasets with different ethnic compositions, and does training on more ethnically diverse data reduce per-group performance disparities?
2. Which geometric facial measurements drive attractiveness predictions across ethnic groups, and how well do interpretable landmark-based features perform relative to end-to-end deep models?

2 Data Profile

2.1 Datasets

We combine three publicly available facial beauty datasets to create a corpus of 17,870 annotated face images, as shown in table 2.1.

SCUT-FBP5500 [2] contains 5,500 front-facing images (2,750 male, 2,750 female) from Asian and Caucasian subjects, each rated by 60 annotators on a 1–5 Likert scale. **MEBeauty** [3] provides 2,370 images across six ethnic groups (Asian, Black, Caucasian, Hispanic, Indian, Middle Eastern) rated by approximately 300 Amazon Mechanical Turk workers on a 1–10 scale, making it significantly more diverse but smaller.

LiveBeauty [4] contributes 10,000 Asian faces with 200,000 total pairwise beauty labels converted to mean scores on a 1–5 scale. Its score distribution is highly concentrated:

Table 2.1: Summary of datasets used.

Dataset	N	Scale	Ethnicities	Raters/Image	Source
SCUT-FBP5500	5,500	1–5	2	60	Kaggle
MEBeauty	2,370	1–10	6	~300	GitHub
LiveBeauty	10,000	1–5	1	~20	Tianchi

45.6% score 3 and 49.8% score 4, with 95.4% of samples in just two bins (see Section 5.7 for implications).

After landmark extraction via MediaPipe Face Mesh and computation of geometric ratios from those landmarks (Chapter 3), the final tabular dataset contains 17,870 rows and 50 columns: 30 geometric beauty markers, 9 expression-related features, and metadata columns (dataset source, ethnicity, gender, raw score, z-normalized score). The extraction success rate was 99.7% (54 images failed).

The combined dataset is heavily imbalanced: Asian faces account for 80.3% (14,351), Caucasian 13.9% (2,480), and remaining groups 5.8% (Black: 296, Hispanic: 296, Middle Eastern: 291, Indian: 156).

2.2 Label Normalization

Since the three datasets use different rating scales (1–5, 1–10, 1–5), we apply **z-score normalization per dataset**: $z_i = (x_i - \mu_d)/\sigma_d$. We considered alternatives: (1) **min-max scaling** to $[0,1]$, which is sensitive to outliers; and (2) **percentile-rank transformation**, which discards distance information. We chose z-scores because they preserve relative distances while aligning center and scale, which is important for regression.

3 Preprocessing and Feature Engineering

3.1 Landmark Extraction and Feature Computation

We use Google’s MediaPipe Face Mesh [1] to detect 478 facial landmarks plus 52 blend-shape coefficients per image. It operates in three steps: (1) face detection (single-face constraint), (2) landmark localization (468 facial + 10 iris landmarks), and (3) blend-shape estimation (52 expression coefficients). All processing runs on CPU, requiring approximately 0.1 seconds per image.

From the 478 landmarks, we compute 30 geometric beauty markers organized into five

categories (Table 3.1). All distances are expressed as ratios to face width or height, eliminating dependence on image resolution. Bilateral symmetry is computed from 9 landmark pairs (e.g., left/right eye corners, nostrils, mouth corners): for each pair we measure the horizontal distance of each landmark from a vertical midline through the nose tip and chin, so that a perfectly symmetric face yields zero left-right difference. `phi_deviation` measures how far the face deviates from the golden ratio ($\varphi = 1.618$). Additionally, 9 expression features derived from blendshapes (smile intensity, frown, jaw openness, etc.) control for expression confounds.

Table 3.1: Summary of extracted feature categories (30 geometric features).

Category	Count	Example Features
Eyes	9	canthal_tilt, eye_width_ratio, eye_area_ratio, eye_asymmetry
Nose	3	nose_width_ratio, nose_length_ratio, nose_symmetry
Mouth/Lips	7	lip_fullness_ratio, mouth_chin_ratio, cupids_bow_ratio
Face & Jaw	7	facial_symmetry, jaw_width_ratio, gonial_angle, chin_taper
Proportions	4	midface_ratio, lower_face_ratio, phi_deviation, facial_thirds_symmetry

3.2 Data Cleaning

No missing value imputation was required since MediaPipe either produces a full landmark set or fails entirely (54 images removed). Features with physically implausible values are capped at the 99.5th percentile, which also serves as a validity check for ratios that should lie within plausible anatomical ranges. We verified that the three datasets contain no duplicate images, and metadata fields (gender, ethnicity) were consistent within each source so no inconsistency resolution was required.

Outlier detection: We flag samples with high annotator disagreement by computing the per-sample standard deviation of rater scores. Samples in the top 10% by standard deviation within each dataset are marked as “high variance.” This information is available for SCUT-FBP5500 and MEBeauty; for LiveBeauty, variance data is largely unavailable ($\approx 90\%$ zero values), so flagging is not applied there.

Binning and feature selection: Binning and explicit feature-subset selection were not applied: the 39 features are continuous geometric ratios where binning would discard precision, and the feature set is small enough that all candidates can be retained for the modeling stage.

4 Data Mining Methods

4.1 Models

We train seven distinct models spanning different algorithmic families, all taking the same 39-feature input vector (30 geometric + 9 expression). As **baselines**, we use a Global Mean Predictor (always predicts the training-set mean z-score, which is close to zero by construction) and **Ridge Regression** (a linear model that separates the contribution of nonlinear interactions from raw linear signal).

We compare three **gradient-boosted tree** variants: **XGBoost** [5] (1,000 trees, max depth 6, learning rate 0.05, early stopping with patience=50), **LightGBM** [6] (leaf-wise growth, 31 leaves, histogram-based splits), and **CatBoost** [7] (ordered boosting with 1,000 iterations). A **Stacking Ensemble** combines four base learners (XGBoost, Random Forest, Gradient Boosting, Ridge) via a Ridge meta-learner, using internal 5-fold CV to generate out-of-fold meta-features. We also train a **3-layer MLP** (128–64–32 neurons, ReLU, Adam optimizer, early stopping on validation loss). A **Quantile XGBoost** variant predicts the median rather than the mean, reducing outlier sensitivity.

4.2 Hyperparameter Optimization and Evaluation

Hyperparameters for gradient-boosted trees are optimized with **Optuna** [8] (TPE sampling, 200 trials) over $\text{max_depth} \in [3, 10]$, $\text{learning_rate} \in [0.01, 0.3]$ (log-uniform), $\text{n_estimators} \in [100, 2000]$, $\text{subsample/colsample_bytree} \in [0.5, 1.0]$, and L1/L2 regularization $\in [10^{-8}, 10]$, minimizing 5-fold CV MAE.

Holdout Test Set: We use a stratified 80/20 train/test split (stratified by ethnicity) with 5-fold CV on the training portion. Within each fold, 10% is held out for early stopping. **Cross-Dataset Generalization:** We train on one dataset and evaluate on the others (6 combinations) to test ethnic transfer. **Fairness:** Using out-of-fold predictions, we compute per-ethnicity and per-gender metrics. Fairness is quantified via the absolute fairness gap ($\text{max group MAE} - \text{min group MAE}$).

Our primary metric is **MAE** (Mean Absolute Error), reported on each dataset’s native rating scale (1–5 for SCUT and LiveBeauty, 1–10 for MEBeauty). We train on z-normalized scores to combine datasets with different scales, but convert prediction errors back to native units for interpretability. We adopt Spearman ρ as the primary correlation metric because it measures ranking ability independent of target distribution shape, making it valid across both continuous (SCUT, MEBeauty) and near-binary (LiveBeauty) targets. Pearson r is reported alongside for comparability with prior literature. As this is a regression task with symmetric error costs, no cost matrix is applied.

5 Results and Discussion

5.1 Overall Model Performance

We evaluate all models against a **Global Mean Predictor** baseline on the combined holdout test set (stratified 80/20 split, 3,574 test samples). Table 5.1 shows correlation metrics and MAE improvement over baseline.

Table 5.1: Model performance on held-out test set (combined across all datasets).

Model	Spearman ρ	Pearson r	MAE Improvement (%) ¹
Global Mean (baseline)	—	—	—
Ridge Regression	0.650	0.660	30.0
Decision Tree	0.538	0.546	22.7
Random Forest	0.648	0.657	27.8
XGBoost	0.700	0.708	35.4
LightGBM	0.697	0.705	35.2
CatBoost	0.701	0.709	35.3
Stacking Ensemble	0.702	0.710	35.4
MLP (Neural Network)	0.706	0.714	36.6
Quantile XGBoost	0.688	0.696	34.7

All learned models substantially outperform the baseline, confirming that the 30 geometric features carry genuine predictive signal. Even Ridge Regression reduces MAE by 30%, showing meaningful linear structure. The gradient-boosted tree family clusters tightly around 35% MAE improvement, while the MLP achieves the largest improvement at 36.6%. Since the differences between MLP, CatBoost, Stacking, and XGBoost are within 1.5 percentage points, we adopt XGBoost as the main model for the per-dataset, cross-dataset, and SHAP analyses that follow: it offers near-best accuracy with substantially lower training cost than the MLP or Stacking ensemble and native compatibility with TreeExplainer SHAP. Table 5.2 reports XGBoost’s per-dataset performance on the native rating scales.

Table 5.2: Per-dataset MAE on native rating scale (XGBoost, 5-fold CV).

Dataset	Scale	MAE	Spearman ρ
SCUT-FBP5500	1–5	0.38	0.76
LiveBeauty	1–5	0.32	0.71
MEBeauty	1–10	0.74	0.65

¹Percentage reduction in MAE relative to the Global Mean baseline, computed in z-score space (baseline MAE = 0.858).

5.2 Cross-Dataset Generalization

Table 5.3: Cross-dataset generalization (train on one dataset, test on another). MAE on test dataset’s native scale.

Experiment	Test Scale	MAE	Spearman ρ	Test N
SCUT $\rightarrow$ MEBBeauty	1–10	1.13	0.29	2,370
SCUT $\rightarrow$ LiveBeauty	1–5	0.44	0.50	10,000
MEBeauty $\rightarrow$ SCUT	1–5	0.53	0.37	5,500
MEBeauty $\rightarrow$ LiveBeauty	1–5	0.43	0.51	10,000
LiveBeauty $\rightarrow$ MEBBeauty	1–10	1.14	0.43	2,370
LiveBeauty $\rightarrow$ SCUT	1–5	0.50	0.52	5,500

The worst generalization occurs for **SCUT $\rightarrow$ MEBBeauty** ($\rho=0.29$), confirming that a model trained on only Asian and Caucasian faces fails when tested on a diverse population (Black, Hispanic, Indian, Middle Eastern subjects), directly addressing research question (1). Transfer between datasets sharing an ethnic group works moderately: SCUT $\leftrightarrow$ LiveBeauty ($\rho=0.50$ – 0.52) both contain majority-Asian faces, suggesting shared geometric patterns within an ethnic group across different rater populations.

5.3 Fairness Analysis: Multi-Model Bias Comparison

To understand whether model choice affects fairness, we compute per-ethnicity MAE for multiple model families and report the **absolute fairness gap** (maximum group MAE minus minimum group MAE) as well as the per-group deviation from the overall MAE.

Table 5.4: Per-ethnicity MAE (z-score) across model families (holdout test set). Gap = max – min group MAE.

Model	Asian	Cauc.	Black	Hisp.	Mid-E.	Indian	Gap
Ridge	0.587	0.616	0.582	0.762	0.743	1.038	0.456
Decision Tree	0.648	0.648	0.689	0.840	0.716	1.007	0.359
Random Forest	0.605	0.635	0.607	0.780	0.686	1.020	0.415
XGBoost	0.552	0.601	0.616	0.717	0.718	0.876	0.324
MLP	0.531	0.575	0.593	0.698	0.703	0.851	0.320

Key observations: (1) The absolute fairness gap ranges from 0.320 (MLP) to 0.456 (Ridge), showing that more expressive models reduce but do not eliminate ethnic disparities. (2) The Indian subgroup consistently has the highest MAE across all models, suggesting that the bias is driven by *data scarcity* (N=156) rather than any particular

model’s inductive bias. (3) The gap between Asian and Indian MAE is approximately 0.3–0.45 z-score units regardless of model family, suggesting that the fairness problem is fundamentally a data problem: no model architecture can compensate for having 92× fewer training examples for Indian faces than Asian faces.

5.4 Feature Importance and SHAP Analysis

Table 5.5: Top-5 features by mean absolute SHAP value, per ethnicity.

Ethnicity	Top-5 Features (ranked)	Top SHAP
Asian	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.264
Black	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.677
Caucasian	nose_width, eye_width, mouth_chin, eye_area, brow_arch	0.263
Hispanic	nose_width, eye_width, mouth_chin, eye_area, brow_arch	0.319
Indian	nose_width, eye_width, mouth_chin, eye_area, jaw_width	0.330
Mid-Eastern	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.318

The top-4 features (**nose width ratio**, **eye width ratio**, **mouth-chin ratio**, and **eye area ratio**) are identical across all ethnic groups, as determined by TreeExplainer SHAP values [9]. Only the 5th feature varies: lip fullness for Asian, Black, and Mid-Eastern faces; brow arch height for Caucasian and Hispanic; jaw width for Indian. The magnitude of influence differs: for Black faces, nose width ratio has 2.6× the SHAP importance (0.677) compared to Asian faces (0.264). This suggests the model relies more heavily on nose geometry to distinguish attractiveness within the Black subgroup, possibly because nose proportions show higher within-group variance for this ethnicity. Notably, **canthal tilt**, despite its popularity in online beauty discourse, ranks 17th–23rd out of 39 features across all groups, suggesting it has minimal predictive value when other geometric features are available. Symmetry-based features (facial symmetry, eye symmetry) show Pearson correlation with attractiveness below $r = 0.04$, likely reflecting MediaPipe’s design: landmark detection places symmetric landmarks by construction, so symmetry features measure detection consistency rather than actual facial asymmetry. Eye asymmetry (difference in left-right eye height) conversely shows mild signal at $r = -0.11$.

5.5 Decision Tree Analysis

As a symbolic, fully interpretable counterpart to the SHAP analysis above, we train a Decision Tree (max depth 7, 124 leaves). Figure 5.1 shows the top three split levels. The root splits on **eye_area_ratio**, separating below-average ($\bar{z} = -0.33$) from above-average ($\bar{z} = +0.38$) faces. Subsequent levels use **eye_width_ratio**, **nose_width_ratio**, and **gonial_angle** – the same features identified as most important by SHAP. Despite lower accuracy ($\rho=0.54$ vs. 0.70 for XGBoost), the tree yields directly readable rules mapping facial proportions to attractiveness.

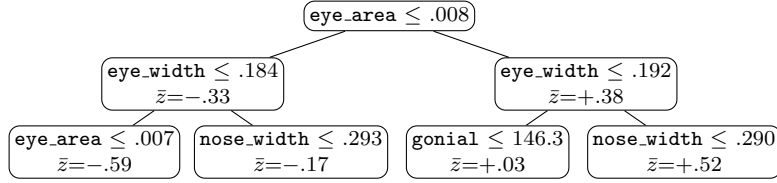


Figure 5.1: Top three levels of the Decision Tree. Left children take the $\leq$ branch.

5.6 Comparison to Benchmarks

5.6.1 vs. Most-Upvoted Kaggle Notebook

The most-upvoted Kaggle notebook [10] for SCUT-FBP5500 trains a 4-layer CNN from scratch on raw 128×128 pixel images (10 epochs, no pretraining, no augmentation), achieving RMSE ≈ 0.51 on a single 80/20 split. Our XGBoost model trained on SCUT-only data achieves RMSE ≈ 0.44 under 5-fold stratified CV, outperforming it by approximately 15% despite using only 30 interpretable geometric measurements rather than raw pixels.

5.6.2 vs. Academic SOTA (FPEM)

The Facial Prior Enhanced Multi-modal model (FPEM) [4] represents the current academic state-of-the-art, combining geometric and appearance features with a deep learning architecture. It benchmarks on all three of our datasets:

Table 5.6: Comparison with FPEM (academic SOTA) by Spearman ρ .

Dataset	FPEM	Ours (XGBoost)	Gap
SCUT-FBP5500	0.931	0.700	-0.231
LiveBeauty	0.925	0.706	-0.219
MEBeauty	0.787	0.648	-0.139
Cross-dataset (LB $\rightarrow$ ME)	0.640	0.418	-0.222

The gap is smallest on MEBeauty (the most ethnically diverse dataset), suggesting that appearance cues help more on homogeneous data where geometric variation is lower. FPEM’s cross-dataset transfer ($\rho=0.64$) also degrades substantially from its within-dataset performance, confirming that cross-cultural generalization remains an open challenge regardless of approach.

5.7 Limitations

Our combined corpus is heavily skewed toward Asian faces (80%), so per-group results for under-represented ethnicities carry wider confidence intervals and the 0.32 z-score

fairness gap is partly a symptom of insufficient training data rather than model bias alone. Compounding this, the attractiveness ratings themselves reflect the demographics of their annotators (predominantly East Asian for LiveBeauty, mixed for MEBeauty), which we cannot fully decouple from subject ethnicity, leaving rater-from-ratee bias as a confound we can flag but not eliminate.

A second cluster of limitations concerns our modeling choices. By design, the geometric pipeline ignores skin texture, color, hair, and grooming, all of which influence human judgments of attractiveness; this exclusion is the most likely explanation for the persistent gap to CNN-based approaches such as FPEM. Relatedly, MediaPipe places left–right landmarks symmetrically by construction, so the symmetry features in our feature set measure landmark-detection consistency rather than true facial asymmetry, which explains their near-zero predictive power. All models also exhibit standard-deviation ratios below 1.0 (0.63–0.85), a regression-to-the-mean effect that compresses the predicted range relative to the true label distribution.

Finally, the LiveBeauty target distribution, with 95.4% of samples concentrated in just two score bins, behaves more like a near-binary classification target than a continuous regression problem, which inflates aggregate accuracy metrics without reflecting genuine fine-grained ranking ability.

6 Conclusion

We trained seven regression models on a corpus of 17,870 faces across 3 datasets and 6 ethnic groups, using 30 interpretable geometric features extracted from MediaPipe landmarks plus 9 expression features.

RQ1 (cross-dataset generalization and fairness). Models trained on ethnically homogeneous data generalize poorly to diverse populations: SCUT (Asian/Caucasian only) transfers to MEBeauty at $\rho = 0.29$, versus $\rho \approx 0.70$ when trained on the combined corpus. Combining heterogeneous datasets improves both accuracy and per-group fairness, but does not close the gap: the absolute fairness gap remains 0.32–0.46 z-score units between best- and worst-served groups across all model families, with the smallest subgroup (Indian, N=156) consistently the worst-served. This indicates the disparity is driven by data scarcity rather than model architecture.

RQ2 (feature drivers and comparison to deep models). The top predictors (nose width, eye width, mouth-chin proportions, eye area) are identical across all six ethnic groups, but their SHAP magnitudes vary (e.g., $2.6\times$ higher for nose width in Black vs. Asian faces), suggesting beauty perception has both universal and culture-specific components. Compared to FPEM, the academic SOTA combining geometry and appearance, our geometry-only approach trails by $\rho \approx 0.22$ on within-dataset evaluation; the gap reflects texture cues (skin, hair, grooming) that landmarks cannot capture, and shrinks on the most diverse dataset (MEBeauty), where geometric variation dominates. Closing this gap would require combining geometric features with texture embeddings (e.g., ArcFace).

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Ehrenwörtliche Erklärung

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Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
Claude Opus 4.6	Implementation of functions we specified; debugging	Codebase	++
Claude Opus 4.6	Wording and structure suggestions on text we authored	Report	+



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Unterschrift

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